

SPIT
(Lecture Notes: Chapter 2)

Chapter 2 : Social Context of Computing

(5 hr)

- 2.1 Society and Technology
 - 2.1.1 Impact of Technology on Society and Vice Versa
 - 2.1.2 Using Technology for Poverty Alleviation
 - 2.1.3 Health Related Issues for an IT Professional
- 2.2 Internet and Society
 - 2.2.1 Digital Divide and Bridging the Digital Divide
 - 2.2.2 Governance of Internet
 - 2.2.3 E-Governance and E-Government Systems

1.0 Society and technology

Definition of Society:

Many definitions of society; the definitions keeps changing, since society is dynamic.

- i. **Schaefer and Lamm:** the largest form of human group, which consists of people who share common heritage and culture. (Richard T. Schaefer and Robert P. Lamm)
- ii. **Ian Robertson:** society gives content, direction and meaning to our lives, and we, in turn, in countless ways, reshape the society that we leave to the next generation. Society is a population that occupies the same territory, is subject to the same political authority and participates in a common culture.
- iii. **P. Gisbert:** a complicated network of social relationships by which every human being is interconnected with his fellowmen.
- iv. **MacIver and Page:** a system of usages and procedures, authority and mutual aid, of many groupings and divisions, of controls of human behavior and of liberties.
- v. **Ginsberg:** a collection of individuals unified by certain relation or modes of behavior, which mark them off from others who do not enter into these relations or who differ from them in behavior
- vi. **A.W. Green:** the targets group to which any individual belongs
- vii. **John F. Cuber:** a group of people who have lived long enough to become organized and to consider themselves and considered as a unit more or less distinct from other human units.

1.0.1 A settlement with the following criteria is called a society

- i. **Population:** A society has population. It can be small or big.
- ii. **Common territory:** The people of a society occupy a common territory.
- iii. **Government or political authority:** The people of a society have a common government and political authority.
- iv. **Common culture and a sense of relationship/membership and committed to group.**

1.0.2 **Essential elements of a society** (RPA, TBG, p1)

- i. People and plurality: society consists of people, of both sexes, of different physical features (age, skin color, eye color, hair color, height, weight, face type)
- ii. Likeness and differences: Members of a society are alike in many aspects. This “alike” bonds them, and they seek understanding and cooperation from each other. The members of a society are different in many ways too, like culture (language, religion, art, music, dance, food, dress, values, belief system), profession, interest, opinion, views etc.
- iii. Cooperation and interdependence: Members of a society cooperate with each other, and they depend on each other. When the level of mutual cooperation reduces, society tends to disintegrate.
- iv. Stable and dynamic: Society is relatively stable; the norms, values, and culture are normally stable/. Yet, society is dynamic. Some elements of a society change slowly, and some change rapidly, depending on external and internal factors.

1.0.3 **Types/Evolution of Society**

- i. Tribal: hunting and gathering society
- ii. Pastoral: domesticate animals
- iii. Horticultural: domesticate plants
- iv. Agricultural/feudal: cultivation of crops, animal energy, irrigation, saving of seed
- v. Industrial: mechanized production, mass production
- vi. Post-industrial: information and service, generation of knowledge

1.0.4 **Factors/Drivers/Causes of social change**

- i. Physical environment/Contact with other societies: Physically easily accessible society changes rapidly than those located in remote (difficult to access) areas due to frequent contacts with members of different societies.
- ii. Natural causes: Earthquake, landslide, flood, desertification, and tsunami disintegrate social fabric and changes society.
- iii. Technological: Information, knowledge and skills: Rate of social change depends on access to information, ability to put together the information into knowledge, ability to convert the knowledge into skills
- iv. Anthropogenic (Human) activities: International War, Civil War, displacement for “developmental” activities, industrial accidents, mass migration, education, economic opportunity

1.0.5 **Classical Theories of social change:**

- i. Cyclical: ups and downs, birth and death, Oswald Spengler: approximate 1000 year cycle. Critics of this theory cite examples that do not follow the 1000 year cycle.
- ii. Evolution: changing with time, getting complex with time, society develops through time and change.
- iii. Functionalist: changes as required to keep the whole society functioning, when particular part of a society changes; based on the assumption that society is a

stable, orderly system; August Comte (1798-1857), Herbert Spencer, Emile Durkheim, and Talcott Parsons propagate this theory. Society is composed of interrelated parts, each of which serves a function and contributes to the overall stability of the society. This theory emphasizes on changing role of different parts of a society to maintain stability of a society.

- iv. Conflict: Karl Marx (19th century) social class conflict between haves and have-nots. Expanded version of conflict theory finds social conflict between any groups in which the potential for inequity exists: racial, gender, religions, political, economic, and so on. Unequal groups usually have conflicting values and agendas, causing them to compete against one another. This constant competition between groups forms the basis for the ever-changing nature of society.

The triad is usually described in the following way:

- The thesis is an intellectual proposition.
- The [antithesis](#) is simply the negation of the thesis, a reaction to the proposition.
- The synthesis solves the conflict between the thesis and antithesis by reconciling their common truths and forming a new thesis, starting the process over.

Cyclical: The Greeks were the first to utilize this model. Plato spoke of eras of time when, initially, hope blossoms only to deteriorate as that era disintegrates. Spengler's view of change was very similar to that of the Greeks, but his model was dressed in a biological rather than uulologica1 garb. Culture, according to him, is the living entity of people, and culture is housed in the civilization of that it arises, develops, ripens, decays and falls never to return. An element of hope, however, was predicted in a similar model by Toynbee, who believed the cyclical change could be interrupted by the creative minority. More recently Sorokin considered that social changes follow a trendless cyclic pattern, i.e., like a swinging pendulum, culture moves in one direction and then back in another.

The functionalist perspective, also called functionalism, is one of the major theoretical perspectives in sociology. It has its origins in the works of Emile Durkheim, who was especially interested in how social order is possible or how society remains relatively stable.

Functionalism interprets each part of society in terms of how it contributes to the stability of the whole society. Society is more than the sum of its parts; rather, each part of society is functional for the stability of the whole society. The different parts are primarily the institutions of society, each of which is organized to fill different needs and each of which has particular consequences for the form and shape of society. The parts all depend on each other.

For example, the government, or state, provides education for the children of the family, which in turn pays taxes on which the state depends to keep itself running. The family is dependent upon the school to help children grow up to have good jobs so that they can raise and support their own families. In the process, the children become law-abiding, taxpaying citizens, who in turn support the state. If all goes well, the parts of society produce order, stability, and productivity. If all does not go well, the parts of society then must adapt to recapture a new order, stability, and productivity.

Functionalism emphasizes the consensus and order that exist in society, focusing on social stability and shared public values. From this perspective, disorganization in the system, such as deviant behavior, leads to change because societal components must adjust to achieve stability. **When one part of the system is not working or is dysfunctional, it affects all other parts and creates social problems, which leads to social change.**

The functionalist perspective achieved its greatest popularity among American sociologists in the 1940s and 1950s. While European functionalists originally focused on explaining the inner workings of social order, American functionalists focused on discovering the functions of human behavior. Among these American functionalist sociologists is [Robert K. Merton](#), who divided human

functions into two types: manifest functions, which are intentional and obvious, and latent functions, which are unintentional and not obvious. The manifest function of attending a church or synagogue, for instance, is to worship as part of a religious community, but its latent function may be to help members learn to discern personal from institutional values. With common sense, manifest functions become easily apparent. Yet this is not necessarily the case for latent functions, which often demand a sociological approach to be revealed.

Functionalism has received criticism for neglecting the negative functions of an event such as divorce. Critics also claim that the perspective justifies the status quo and complacency on the part of society's members. Functionalism does not encourage people to take an active role in changing their social environment, even when such change may benefit them. Instead, functionalism sees active social change as undesirable because the various parts of society will compensate naturally for any problems that may arise.

References

Anderson, M.L. and Taylor, H.F. (2009). Sociology: The Essentials. Belmont, CA: Thomson Wadsworth.

CliffsNotes.com. Three Major Perspectives in Sociology. 22 Jun 2011.

http://www.cliffsnotes.com/study_guide/topicArticleId-26957,articleId-26837.html.

Source: <http://sociology.about.com/od/Sociological-Theory/a/Functionalist-Theory.htm>

Accessed: June 14, 2013

Conflict theory emphasizes the role of coercion and power in producing social order. This perspective is derived from the works of Karl Marx, who saw society as fragmented into groups that compete for social and economic resources. Social order is maintained by domination, with power in the hands of those with the greatest political, economic, and social resources. When consensus exists, it is attributable to people being united around common interests, often in opposition to other groups.

According to conflict theory, inequality exists because those in control of a disproportionate share of society's resources actively defend their advantages. The masses are not bound to society by their shared values, but by coercion at the hands of those in power. This perspective emphasizes social control, not consensus and conformity. Groups and individuals advance their own interests, struggling over control of societal resources. Those with the most resources exercise power over others with inequality and power struggles resulting. There is great attention paid to class, race, and gender in this perspective because they are seen as the grounds of the most pertinent and enduring struggles in society.

Whereas most other sociological theories focus on the positive aspects of society, conflict perspective focuses on the negative, conflicted, and ever-changing nature of society. Unlike functionalists who defend the status quo, avoid social change, and believe people cooperate to effect social order, conflict theorists challenge the status quo, encourage social change (even when this means social revolution), and believe rich and powerful people force social order on the poor and the weak. Conflict theorists, for example, may interpret an "elite" board of regents raising tuition to pay for esoteric new programs that raise the prestige of a local college as self-serving rather than as beneficial for students.

Whereas American sociologists in the 1940s and 1950s generally ignored the conflict perspective in favor of the functionalist, the tumultuous 1960s saw American sociologists gain considerable interest in conflict theory. They also expanded Marx's idea that the key conflict in society was strictly economic. Today, conflict theorists find social conflict between any groups in which the potential for inequality exists: racial, gender, religious, political, economic, and so on. Conflict theorists note that unequal groups usually have conflicting values and agendas, causing them to compete against one another. This constant competition between groups forms the basis for the ever-changing nature of society. Critics of the conflict perspective point to its overly negative view of society. The theory ultimately attributes humanitarian efforts, altruism, democracy, civil rights, and other positive aspects of society to capitalistic designs to control the masses, not to inherent interests in preserving society and social order.

References

Anderson, M.L. and Taylor, H.F. (2009). Sociology: The Essentials. Belmont, CA: Thomson Wadsworth.

CliffsNotes.com. Three Major Perspectives in Sociology. 22 Jun 2011.

http://www.cliffsnotes.com/study_guide/topicArticleId-26957,articleId-26837.html.

Source: <http://sociology.about.com/od/Sociological-Theory/a/Conflict-Theory.htm>

Accessed: June 14, 2013

There are many theories of social change. Generally, a theory of change should include elements such as structural aspects of change (like population shifts), Processes and mechanisms of social change, and directions of change.

- **Hegelian:** The classic [Hegelian](#) dialectic model of change is based on the interaction of opposing forces. Starting from a point of momentary stasis, Thesis countered by Antithesis first yields conflict, then it subsequently results in a new Synthesis.
- **Marxist:** [Marxism](#) presents a dialectical and materialist concept of history; Humankind's history is a fundamental struggle between [social classes](#).
- **Kuhnian:** The [philosopher of science](#), [Thomas Kuhn](#) argues in [The Structure of Scientific Revolutions](#) with respect to the [Copernican](#) Revolution that people are unlikely to jettison an unworkable paradigm, despite many indications that the paradigm is not functioning properly, until a better paradigm can be presented.
- **Heraclitan:** The Greek philosopher [Heraclitus](#) used the metaphor of a river to speak of change thus, "On those stepping into rivers staying the same other and other waters flow" (DK22B12). What Heraclitus seems to be suggesting here, later interpretations notwithstanding, is that, in order for the river to remain the river, change must constantly be taking place. Thus one may think of the Heraclitan model as parallel to that of a living organism, which, in order to remain alive, must constantly be changing. A contemporary application of this approach is shown in the social change theory [SEED-SCALE](#) which builds off of the [Complexity Theory](#) subfield of [Emergence](#).
- **Daoist:** The Chinese philosophical work [Dao De Jing](#), I.8 and II.78 uses the metaphor of water as the ideal agent of change. Water, although soft and yielding, will eventually wear away stone. Change in this model is to be natural, harmonious and steady, albeit imperceptible.
- **Resource-based economy:** [Jacque Fresco](#)'s concept of a resource-based economy that replaces the need for the current monetary economy, which is "scarcity-oriented" or "scarcity-based". Fresco argues that the world is rich in natural resources and energy and that — with modern technology and judicious efficiency — the needs of the global population can be met with abundance, while at the same time removing the current limitations of what is deemed possible due to notions of economic viability. Fresco's work in [The Venus Project](#) deals with physical reality and natural law rather than economic abstractions. [The Venus Project](#)'s website says this: "The Venus Project is neither Utopian nor Orwellian, nor does it reflect the dreams of impractical idealists. Instead, it presents attainable goals requiring only the intelligent application of what we already know."¹

Source:

https://en.wikipedia.org/wiki/Social_change#Prominent_theories_of_social_change

Accessed: June 14, 2013

1.0.6 Eastern and Western values of society

Eastern Values

Panchasheel:

1. Mutual respect for each other's territorial integrity and sovereignty,
2. Mutual non-aggression,
3. Mutual non-interference in each other's internal affairs,
4. Equality and mutual benefit, and
5. Peaceful co-existence

- i. Social Harmony
- ii. Sacrifice for the welfare of group
- iii. Modesty and frugality
- iv. Truth and Integrity
- v. Respect to elder, teacher, guest and ancestor
- vi. Maintaining culture and tradition
- vii. Helping the people in need (Paropakar)
- viii. Purity in thought and acts
- ix. Ritual as important as devotion or work

Western/American Values (Robin Williams, 1970)

- i. Achievement and success
- ii. Activity and work
- iii. Moral orientation
- iv. Humanitarianism
- v. Efficiency and practicality
- vi. Progress
- vii. Material comfort
- viii. Equality
- ix. Freedom
- x. External conformity
- xi. Science and rationality
- xii. Nationalism-patriotism
- xiii. Democracy
- xiv. Individual personality
- xv. Group-superiority themes

1.1 Impacts and Consequences of Technology on Society

1.1.1 Socio-cultural Impact

Impact on	Impact
Livelihood	Livelihood is diversifying, many types of livelihood options available
Specialization of Profession	Specialization in profession is increasing.
Health	Access to health services is increasing, specific medicines are available, diagnosis process improving, genetic engineering is curing previously incurable diseases, assistance to specific needs of disabled persons getting better through technically designed equipment, welfare of aged citizens getting better
Education	Access to education services getting better, distance learning is improving access to education and latest information, e-book, e-library, standardization of educational quality through use of ICT technology, on line evaluation, on line test possible,
Family	Family relation getting complex due to in-vitro fertilization, test-tube baby, cloning, Family size reducing due to easier access to contraception, micro-family getting possible, life span increasing, several generation in family, change in status of female and disabled
Living Standard	Living standard increasing, HDI increasing, poverty decreasing, employment opportunity increasing, price of goods decreasing
Language	Use of international language increasing, language use getting standardized through mass media, brail script helping blind, sign language use increasing for deaf
Social Norms and Values	Social norms and values increasingly being challenged and altered or replaced, social class disintegrating, new economic class emerging
Family tradition, culture, heritage	Family traditions increasingly being challenged and altered or replaced, heritage preservation getting better
Urbanization	Urbanization increasing, concentration of resources and resource users
Commercialized Recreation	Traditional dances, music, drama and games giving ways to movies, video games, theme parks, pay per view programs

1.1.2 Impact on Agriculture

Impact on	Impact
Food Production	Food production mechanized, food production increasing, Green Revolution
Food Processing	Food processing getting complex, requiring industrial management
Food Preservation	Food security increasing
Mechanization	Mechanization of all aspects of agriculture is increasing
Commercialization	From subsistence to cash crop, genetically modified food, patented seeds
Food Variety	Increasing, year round availability
Fertilizer/Pesticide	From organic to chemical, pesticide use increasing
Water use	Increasing due to cash crops, reduced use by sprinkler, drip, root zone

1.1.3 Impact on Communication

Impact on	Impact
Information generation and dissemination	Increasing access to information
Mass communication: Newspaper, Magazines, FM Radio, TV, instant/breaking news	Enhanced public awareness, timely information to people
Internet and Social Media	Increased access to information, social media influencing design of communication techniques
Telephone, mobile phone	Increased and easier access to telephone
Virtual Meeting, virtual reality, tele-medicines	Less need of physical presence in meeting, distance medicine services and remote controller operation possible

1.1.4 Impact on Industry, Economy and Job

Impact on	Impact
Industrial Production and mass production	Increased production of similar size and design
Price of goods	Relatively decreasing due to mass production
Capital intensive technologies and labor saving device use	Increasing
Mechanization, Automation, and Robotics use	Increasing efficiency of industry, economic indicators improving
Job opportunities	Increasing, also for women, after WW II

1.1.5 Impact on Transportation: mass transportation of goods and services

Impact on	Impact
Land transportation	Road, multilevel roads, railways
Underground	Metro/Underground railways
Water	Naval transportation, river navigation
Air	Air cargo, air travel, air ambulance

1.1.6 Impact on Politics

Impact on	Impact
Spreading of message/manifesto and propaganda	Increasing speed spreading political messages and messages reaching to more people
TV Debate, recognition of candidates, election management	People get chance to evaluate the candidates, election managers manipulate media to their advantage
Link between politicians & people	Increasing without physical contact/presence
Social media influencing policies	Increasing
e-voting, e-governance	Increasing

1.1.7 Impact on Dispute/Conflict Resolution:

Impact on	Impact
Warfare	Increased use of gun powder, rockets, missiles, drones, improvised explosive devices
Weapons of mass destruction	atom bomb, hydrogen bomb, nuclear bomb made but not yet used (except two in WW II), Star War, MAD policy
Weapons of mass deception	Misuse/abuse of mass media increasing

1.1.8 Commercialized recreation

Impact on	Impact
Traditional recreation	Declining
Modern recreation	Increasing, Theme parks, swimming pools, video game parlor
International games	Increasingly being costly and commercialized, pay per view, exclusive right to broadcast

1.1.9 Consequences:

- Global Village through development of Transportation technology and ICT technology
- Industrialization and fossil fuel use causing global warming and increase in frequency of extreme events (Global Climate Change)
- Human Development Index parameters improving: GNP, GDP, Per Capita Income, Per Capita Energy use, access to education/health, WATSAN/WASH, employment, poverty, infant mortality

1.2 Effects of Major Technological Developments such as Printing, Gunpowder, Mechanization, Computer, Organic Chemistry, Communication Satellites

1.2.1 Printing:

Written and mass produced record of agreements, laws, rules, regulations, guidelines, procedure, manuals, books, newspapers, magazines, journals, easily available

Effects:

Need to memorize diminished, information available when needed

Judgment based on written laws rather than wisdom of justices,

3D printing: can change production mode, 3D Printers print house:

<http://www.sciencealert.com.au/news/20141510-26336.html>

1.2.2 Gunpowder:

Explosives, Dynamite, Warfare, Colonization, Endangered species, Crime, Terrorism

Effects: warfare getting increasingly violent and costly, species getting extinct, increasing use of gunpowder in crimes and terrorist activities

1.2.3 Mechanization:

Industrial production, agricultural mechanization, transportation, robotics, automation,

Effects:

Lower cost of goods and food

Easier movement over long distances

Standardization and interchangeability of design, size, shape

Safer working environment for dangerous jobs

Higher living standard, material comfort

1.2.4 Computer:

Modeling, simulation (visualization of potential impacts and solution before the event occurs and solutions implemented), design, solution techniques (FDM, FEM), Management Information System, Spatial planning using GIS, computer crimes, intrusion on personal privacy, data mining, data theft,

Effects: information generating, storing and dissemination

Design of materials using simulation

Electronic equipment

Impact of Computer on Society

- Wide application: applicable almost everywhere
- Employment: increased for persons with new skills
- Productivity and competition: increased
- Standard of living: has improved.
- Privacy: reduced

Computer Crimes

Electronic Transaction Act 2004 has listed the following acts under the Computer Crime

- i. Theft, damage or alteration of computer source code
- ii. Unauthorized access to materials on computer
- iii. Destruction and computer and information system
- iv. Publication of illegal materials on electronic form
- v. Violation of confidentiality
- vi. Posting of false information
- vii. Submitting or showing forged license or certificate
- viii. Non-submission of the document or other materials
- ix. Computer fraud

1.2.5 **Organic Chemistry:**

Plastic, Polymer, plastic, Construction glues, fossil fuel, construction materials

Effects: New materials: including construction materials, PPR pipes, PVC pipes, synthetic glues, synthetic clothes, bio-degradable plastic, light weight and stronger materials, water proofing, paint

Fossil fuel: transportation, bitumen

1.2.6 **Communication Satellites:**

Mass Communication: Radio, TV, Newspaper, Internet, WWW, social media, virtual meetings, Satellite maps, GIS, GPS, Map making, Open Source Mapping, Google Maps, Global Circulation Modeling, Surveying, Wild Fire detection, Early Warning

Effects: ICT development, TV, GPS, Satellite Maps, Weather forecasting, Climate Modeling, LiDAR, GIS based planning (urban, land use etc), Remote Sensing

2.1.2 Using Technology for Poverty Alleviation

- The figure of US\$1 income per day is widely accepted as a general indicator of extreme poverty.
- There is no absolute cut-off and income is only one indicator of the results of poverty, among many others.
- According to the World Bank, poverty includes powerlessness, voicelessness, vulnerability, and fear.
- Poverty also include the deprivation of basic capabilities and lack of access to education, health, natural resources, employment, land and credit, political participation, services, and infrastructure (European Commission, 2001).

The ICT-Poverty connection

- A knowledge gap is an important determinant of persistent poverty.
- Developed countries already possess the knowledge required to assure a universally adequate standard of living.
- ICT can encourage greater interaction and communication within and between countries in order to reduce the “digital divide”.
- Consider Kofi Anan’s quote on the digital divide as an indicator of poverty.

“The new information and communications technologies are among the driving forces of globalisation. They are bringing people together, and bringing decision makers unprecedented new tools for development. At the same time, however, the gap between information ‘haves’ and ‘have-nots’ is widening, and there is a real danger that the world’s poor will be excluded from the emerging knowledge-based global economy” (Anan, 2002).

Specific ICT Applications for poverty alleviation

Radio

- Radio has achieved impressive results in the delivery of useful information to poor people. Its principal attribute is its ubiquity.
- A survey of 21,000 farmers enrolled in radio-backed farm forums in Zambia found that 90 percent found programmes relevant and more than 50 percent credited the programmes and forums with increasing their crop yields (Dodds, 1999).
- In South Africa, clockwork radios that do not require battery or electricity are being distributed to villages to enable them to listen to development programming.
- The radio has been used extensively to disseminate health information, weather forecast and so on.

Television

- Not as common as the radio but has been useful in some settings.
- Probably the most notable example of TV for development comes from China with its TV University and agricultural TV station.
- In Việt Nam, two universities in the Mekong Delta Region work with the local TV station to broadcast weekly farmer’s workshops that are watched by millions.

Telephone

- With the introduction of the GSM, there has been a massive expansion of telephone services in the rural areas where the poor live.
- A good example is the Grameen hand phones in Bangladesh, in which the Grameen Bank, the village-based micro-finance organization, leases cellular mobile phones to successful members.
- This has delivered significant benefits to the poor. The phones are mostly used for exchanging price and business and health related information.

- In Nigeria, it is quite obvious the economic benefits of the GSM. Think of how much employment and income it has generated.
- In the rural areas, the phones offer additional non-economic benefits such as improved law enforcement, reduced inequality, more rapid and effective communication during emergencies and stronger kinship bonding.
- In China, villages that had the telephone, the most basic communications technology, experienced declines in the purchase price of various commodities and lower future price variability.
- Village telephones facilitated job searches, improved access to emergency medical care and the ability to deal with natural disasters; lowered mortality rates for livestock due to more timely advice from extension workers; and improved rates in foreign-exchange transactions (Eggleston et al., 2002).

Public address systems

- Commonly used in China and Việt Nam where to deliver public information, announcements and the daily news.
- PA systems are more localized than radio, but are technically simpler and less expensive.
- Research on poor communities indicate that the telephone and radio remain the most important (direct access) ICT tools for changing the lives of the poor (Heeks, 1999).

Computers and the Internet

- Computers and the Internet are commonly made available to poor communities in the form of community-based telecentres.
- These centers provide shared access to computers and the Internet and are the only realistic means of doing this for poor communities. A few case studies will be cited later.
- The two key elements of telecenters are public access and a development orientation.
- It is the latter characteristic that distinguishes telecentres from cyber cafés.
- Telecentres can provide a range of ICT-based services from which they can earn an income, such as telephone use, photocopying and printing, email and word processing.
- This helps with financial self-sustainability, which telecentres are often required to attain.
- Some argue that ICT-based development services should not have to be paid for by poor people, and should be provided as a public service, rather like libraries.

Poverty Alleviation using ICT

- “The most effective route to achieving substantial benefit with ICTs in development programs is to concentrate on re-thinking development activities by analyzing current problems and associated contextual conditions, and considering ICT as just one ingredient of the solution”- Roger Harris.

Successful use of ICT for poverty alleviation

1. Distributing locally relevant information.

- Where information is provided in the local language and with locally relevant content, the community can benefit significantly.

-The Village Information Shops in Pondicherry, India, provides information using the Tamil language and Tamil script in the computers (Sentilkumaran and Arunachalam, 2002).

The above and other similar projects in India provide information on

- | | |
|--|----------------------------------|
| - Commodity marketing information system | - Below-Poverty-Line Family List |
| - Landholder's passbook of land rights and loans | - Employment news |
| - Rural Hindi e-mail | - Rural matrimonial |
| - Forms of various government schemes. | - Rural market |
| | - Rural newspaper |
| | - E-education |

2. Targeting disadvantaged and marginalized groups

- - ICT has been used to benefit disadvantaged groups such as women, disabled, non-English speakers and so on.
- - Such groups usually require special assistance and attention in if they are to benefit from programs that are targeted at poor people.
- - Strategies for reaching marginalized sectors of society through ICTs include the collection, classification, protection, and commercialization of indigenous knowledge by minority groups using ICTs. Traditional remedies are being recorded in databases and afforded protection from foreign applications for patents.

3. Promoting local entrepreneurship

- ICTs have the potential to impact the livelihood strategies of small-scale enterprises and local entrepreneurs in the following areas:
 - Natural capital - opportunities for accessing national government policies
 - Financial capital - communication with lending organizations, e.g., for micro-credit
 - Human capital - increased knowledge of new skills through distance learning and processes required for certification
 - Social capital - cultivating contacts beyond the immediate community
 - Physical capital - lobbying for the provision of basic infrastructure

4. Improving poor people's health

Health care is one of the most promising areas for poverty alleviation through the use of ICTs. This can be achieved in the following areas.

- a. to facilitate remote consultation, diagnosis, and treatment.
- b. To enable health workers in developing countries are accessing relevant medical training through ICT-enabled delivery mechanisms.
- c. Radio, TV (and more recently the Internet) have a history of effectively facilitating the dissemination of public health campaigns and disease prevention strategies in developing countries.

5. Strengthening education

- In developing countries, distance education programmes help to educate more people for less money.
- UNESCO and the World Bank have reported that in the world's 10 biggest distance education institutions, the majority of which are in the Third World, the cost of education per student is on average about one third the cost at traditional institutions in the same country.
- China Central Radio and Television University has 1.5 million students, two-thirds of them in degree programmes.
- The university caters to working adults. It broadcasts radio and TV lectures at fixed times to students at 2,600 branch campuses and 29,000 study centres, as well as at workplaces.
- In primary and secondary education, radio and television are increasingly important means of reaching the rural poor. In Mexico, over 700,000 secondary-school students in remote villages now have access to the *Telesecundaria* program, which provides televised classes and a comprehensive curriculum through closedcircuit television, satellite transmissions and teleconferencing between students and teachers.

6. Promoting trade and e-commerce

- E-commerce is spreading most quickly among developing countries.
- M-commerce, defined as the buying and selling of goods and services using wireless handheld devices such as mobile telephones or personal data assistants (PDAs), is likewise growing at a rapid pace.

- The main areas of m-commerce use are in text messaging or SMS (short messaging service), micro-payments, financial services, logistics, information services and wireless customer relationship management.

8. Supporting good governance

- E-governance is an area of ICT use that shows rapidly increasing promise for alleviating the powerlessness, voicelessness, vulnerability and fear dimensions of poverty.
- ICTs have been used to spread democracy and include the poor in the process of governance.

9. Building capacity and capability

- Capacity building refers to developing an organization's (or individual's) core skills and capabilities to help it (him/her) achieve its (his/her) development goals.
- ICTs can help to achieve this.

The Village Information Shops in Pondicherry, have used ICTs to build awareness in poor communities of the government programs and entitlements that are available for their assistance

10. Enriching culture

- ICTs can simultaneously be a threat and an opportunity to a culture.
- ICTs can help to preserve indigenous culture.
- Aside from digitization of indigenous cultural artefacts, ICTs provide a means for cultural communities to strengthen cultural ties.
- For example, the Internet is helping to unite Assyrian communities, regardless of their geographic, educational, and economic backgrounds.

11. Supporting agriculture

- ICTs can provide useful information to farmers in the area of crop care and animal husbandry, fertilizer and feedstock inputs, drought mitigation, pest control, irrigation, weather forecasting, seed sourcing and market prices.
- ICTs can also enable farmers to participate in advocacy and cooperative activities.

12. Creating employment opportunities

- Unemployed people can use ICTs to discover job opportunities.
- They can become employed in the new jobs that are created through the deployment of ICTs.
- Through open job seeker banks, for example, employers can search and directly access résumés, which in turn are linked electronically to job vacancy banks.
- One common option is to purchase a mobile phone through a micro credit program and to earn income by providing low cost phone calls to others. This is quite common in Nigeria.
- India has seen rapid growth in cyber kiosks that provide access to social communication as well as business support services for underprivileged groups.
- It has helped to reduce youth unemployment.

Limitations of ICT

- ICTs alone are insufficient for significant benefits to emerge.
- ICTs will not transform bad development into good development, but they can make good development better.
- Effective applications of ICTs comprise both a technological infrastructure and an information infrastructure.
- In rural settings in developing countries (where the vast majority of poor people live), it is always a challenge to install the technological infrastructure, but the task is relatively simple compared to establishing the information infrastructure.
- While ICTs provide opportunities for development, desirable outcomes always arise from the actions of people.

Health Related Issues for an IT Professional

A study was done among 200 IT professionals to evaluate the computer related health problems and role of ergonomic factors. The result revealed that there was approximately 93% of computer related morbidity in the subjects. The visual problems were noticed in 76 percent and musculo-skeletal in 77.5 percent. Furthermore, 35 percent faced stress.

The proper use of gadgets and devices that keep health intact is very important, especially when you deal with the same devices for hours together and for years. We call it ergonomics! The ergonomics is also a great concerned about HCI –the Human Computer Interaction (also called as Man Machine Interaction –MMI or Computer Human Interaction –CHI) that involves the planning and design of the computers for better human use.

What is wrong with computer use?

Nothing wrong! They are there to help us but then we forget about health, about ourselves. Let's look at some important points to get into the topic...

The posture –how you sit in front of the computer

The distance –how far is computer from your eyes

The level –at what eye level the computer is fixed

Lack of motion & repetition –the body movement while using computers for the whole day

The breath –the air you breathe in

Internet –how close you are with this buddy

The Health Issues Due to Posture

Back Pain or Low Backache –the lower portion of back pains. The problem becomes severe that you cannot bend forward. This is typically noticed when you get up from your seat after sitting for long hours. Your body feels stiffened and you take a few minutes to get back into your flexible movement.

Poor posture leads to multiple health problems

Remember: Keep Some Meta-stuff to Support Back!

Prevention: The posture does matter when it is sitting in front of the computer. Just check out the curve you make from the lower back. Lesser the curve you make, better the posture it is. Try to sit erected and don't bend forward to type or to see the monitor closely. You can also put some support to fill the curve-gap to prevent such harm. Regular massage helps.

Health Disorders Because of the Distance

Eye strain, dark circles and redness –the eyes are most delicate part of our body. If there is less distance between eyes and the monitor, the rays coming from it will affect your eyes badly.

Computer Vision Syndrome or Dry Eye Syndrome –The eyes are red, itchy and constantly irritating. The simple reason is the screen, its radiation and resolution (i.e. Brightness and contrast). If the problem is severe, very

small particles will be seen where eyes are moved. It's actually IN your eyes and not actually present. Use anti-glare lens/eyeglasses.

Remember: *Be Close to Code but Keep Distance with Monitor!*

Prevention: Ideally, the distance should be 18-24 inches to keep your eyes healthy. Take off the keyboard from the computer desk and arrange it somewhere nearer to you. You do this with mouse too. In case you have enough space on your desk, put the computer screen farthest possible. In CRV monitors, we had safety film/screen but with LED/LCD, it has gone. No matter how great claims you hear from a salesman, radiation from the screen DOES affect your eyes.

Have enough lighting in the room, don't operate computer in dark or poor light condition. The light should fall on the screen and not on your eyes. In other words, the window (if any) should be behind you and not in front. Keep blinking. Look at the objects or scenery that is at long distance every half an hour. Do palming (gently put your palms onto the eyes with a little pressure, concentrate in the center of the dark. Do this for 3-4 minutes). Keep eye lubricants with you and hydrate your eyes every 3-5 hourly. *Bhramari Pranayama* is very useful for this.

Computer Related Health Problems Because of the Monitor Level

Neck pain due to monitor level

Cervical Spondylitis (neck pain) –this happens due to improper level of the computer screen and your eyes. Constant looking low at the monitor will cause this problem as it'll stiffen the muscles of the neck and make them rigid.

Remember: *Customize Applications Only After Customizing Monitor Level!*

Prevention: Adjust the level of the computer monitor and maintain the viewing angle. If this is not possible, adjust your chair –make it higher or shorter with the help of level provided. Move your neck up-down for a several times after every hour. Rotate your head 10 times from end-to-end after every three hours. Regular massage helps

Major Health Problems Due to Lack of Motion

- a) **Constipation & Piles** –it's hard stool in layperson's language. The intestines push the food ahead with their special movements called peristalsis. If you sit for longer hours daily for years, the intestine becomes sluggish. It is also due to improper timing of the meals that is observed in the IT professionals. The intestines then give up their work or become too slow. This results in the indigestion and forming of hard stools and piles too. Needless to say, when the stool remains your intestines for longer duration, the filthy material causes a lot of infection.

Remember: *'Run' for Program but 'Move' for You!*

Prevention: Take your meals on time. Take a walk-break rather coffee break! Practice short spans of Anulom-Vilom Pranayam. Include fiber in your diet (dark, green and leafy vegetables) and avoid too

oily and junk food. After every hour, simply stand up, stretch your body, press your stomach gently and sit down.

- b) **Heart Problem** – According to NPR, a 2010 study revealed that the men with more than 23 hr./week of sedentary lifestyle were 64% greater risks of dying from cardiac arrest as compared to those who with less than 11 hr./week. For IT guys, this is alarming. The increase in cholesterol in IT professionals could be due to improper food timing, junk food, lack of motion and exercises. Improper breathing, chronic constipation, and poor blood circulation can also cause this issue.

Prevention: Remain active. Regularly go for the jogging as cardio exercises are the best to prevent this problem. Practice *Pranayam* and do *yogasanas* as even modern health care practitioners too believe these as very effective holistic techniques to prevent heart disease. Prefer stairs over elevators and park your vehicle a bit far from the parking place so that you walk down to the office.

Obesity in IT professionals

- c) **Obesity or overweight** –this is typical problem that IT professionals face these days. The reason is simple again, lack of motion, activities and exercises. Constant sitting on the chair in front of the computer add extra pounds to your body.

Prevention: Avoid junk food. Stop munching every time and if you cannot, keep sprouts, salads (carrots and cucumbers are best for this) or something fiber rich edible. Regular exercise, *Pranayam*, *yogasanas* and jogging will be great for this. If you are figure conscious, prefer manual mesotherapy (over liposuction) that gives you instant inch loss from targeted body areas.

Carpal Tunnel Syndrome

- d) **Carpal Tunnel Syndrome** –We have tunnel in our wrists! This tunnel consists of muscles, nerves, blood vessels etc. this is due to continuous use of keyboard and mouse. You move your fingers and hand in particular direction several hundred times a day. Stiffness and pain in the wrist, numbness in the fingers and tingling sensation in the hand are few of the symptoms.

Prevention: Rotate your wrist every hour; flex and relax the fingers several times. Put something spongy under your wrist. Keep thin mouse-pad so that the thickness of the pad does not hurt your wrist while moving the mouse. Adjust the level of your sitting. The wrist should be parallel to the desk so that you don't form any angle while typing or moving the mouse.

- e) **eThrombosis** –Deep vein thrombosis; in other words, the clots deep in the legs. This happens due to lack of motion in the legs. The legs bear heavy flow of the blood because of the gravitational force. This major health problem in IT professional is due to long sitting hours on the computer without any physical movement of the legs. The vein in later stage become stiffened, start paining and then gives unbearable sprain and rigidity in the legs.

A 32-year-old person lost his consciousness after a few weeks suffering a swollen calf. A massive clot caused as it moved from veins in his leg to the lungs, what we call a pulmonary embolism. The experts

found no such risk factor for the clot, except long hours (more than 12 hr. a day) of sitting on the computer².

Prevention: Move your legs. Regularly stand up, take a short walk and come back. Do not keep water bottle with you. When feel thirsty, walk down to the water fountain. Jogging, cardio exercises, sports, aerobics etc. are helpful. Regular massage also helps.

Several Major Health Issues Because of Internet Addiction

Typically, Internet addiction is defined as uncontrolled and harmful use of Internet that comes in the forms of sexuality, gaming and excessive use of emails, chats or SMS. There are many studies revealed that abuse of alcohol and tobacco, depression and other major health problems happen due to Internet addiction. In boys and



men depression could be more a result of the addiction than a cause³.

Internet Addiction

Internet addiction brings a lot of health hazards due to over computer and handheld (internet enabled) devices. While enjoying the exemption from social stigmas, the internet users find themselves more comfortable on the imaginary World Wide Web than this vale of tears.

Common problems include depression, dependency, anxiety, Obsessive Compulsive Disorder (OCD), complex problem, stubbornness, lack of social responsibilities, disrespect toward the elders (especially in children) and mood swings. Typically the office-goers feel discomfort on weekends wherein they might not access the internet and start thinking all time about the emails to reply on Monday.

Remember: There is a CTRL Button Even on the Keyboard!

Prevention: Keep your kids' computers in the public hall and periodically check the history of webpages they visit in your absence. Fix up something like Team Viewer or simply block certain addictive websites. But in my opinion, this requires more mental motivation and counseling. If you are a parent, try to spend time with your kids and make them realize that the company of good humans is always better than idiot boxes and funky gadgets. It's always better to introduce such gadgets as a source of knowledge than entertainment. Let them learn the later part later in their life!

If you are an office-goer or someone who is WWW conscious, meditation is very useful. You can alternatively try some The Nature's sound track or some classical music to make your mind stronger leaving the addiction.

Digital Divide

The digital divide is the gap between those countries that receive information by making effective use of computers and other forms of information technology, and those countries that do not socially or economically profit from technological advancements, such as worldwide communication via the web or e-commerce.

Why Bridge the Digital Divide?

Developing countries and successful, industrialized nations can both benefit economically and socially from the Internet expanding and increased worldwide usage of the net. The Internet acts as a gateway that links millions of people use to communicate ideas and act as potential buyers and sellers in a global economy. The Internet has created a worldwide forum for dialogue and has generated a revolution of innovation and entrepreneurship through E-commerce.

Effects of Establishing Equality and Bridging the Gap: Globalization that enables less developed countries to:

- | | |
|---|--|
| a. Acquire knowledge | h. Free access to information |
| b. Enhance educational and teaching methods | i. Enables citizens and businesses in every country, especially developing nations, to make more informed and educated decisions concerning international and national issues. |
| c. Improve government policy-making | j. Potential decrease for terrorism due to increased international dialogue |
| d. Promote sanitation standards and increase health care provisions | k. Stimulates a growth of US political capital and spreads democratic ideals |
| e. Establish a global community | |
| f. Attract a wider range of investors and tourists | |
| g. International E-commerce | |

Effects of Maintaining the Current Level of Inequality

- a. Modernization for some nations
- b. Worldwide stagnation of some foreign markets due to lack of technological tools and an inability to compete economically on an international scale
- c. Limited information access
- d. Limited international dialogue and more restrictive governments in less developed countries that lack Internet access
- e. Potential for increased terrorism due to socioeconomic stress

A continuation of the digital divide could separate the world into two distinct groups: those with the Internet and those countries not able to benefit from technological advancement. Potentially, this divide could spawn a revolution in less developed countries, in which these countries would demand a redistribution of information and technological resources.

There are a variety of arguments regarding why closing the digital divide is important. The major arguments are the following:

1. Economic equality

Some think that the access to the Internet is a basic component of civil life that some developed countries aim to guarantee for their citizens. Telephone is often considered important for security reasons. Health, criminal, and other types of emergencies might indeed be handled better if the person in trouble has an access to the telephone. Another important fact seems to be that much vital information for people's career, civic life, safety, etc. are increasingly provided via the Internet. Even social welfare services are sometimes administered and offered electronically.

2. Social mobility

Some believe that computer and computer networks play an increasingly important role in their learning and career, so that education should include that of computing and use of the Internet. Without such offerings, the existing digital divide works unfairly to the children in the lower socioeconomic status. In order to provide equal opportunities, governments might offer some form of support.

3. Democracy

Some think that the use of the Internet would lead to a healthier democracy in one way or another. Among the most ambitious visions are that of increased public participation in elections and decision making processes.

4. Economic growth

Some think that the development of information infrastructure and active use of it would be a shortcut to economic growth for less developed nations. Information technologies in general tend to be associated with productivity improvements. The exploitation of the latest technologies may give industries of certain countries a competitive advantage.

3 'C' to bridge the digital divide

- Connectivity
Connect the whole world in a network so that everyone can access the information/ contents
- Contents
Shared knowledge is double knowledge and knowledge is power
- Capacity
Gradually increase the capacity

Internet governance

Internet governance is the development and application of shared principles, norms, rules, decision-making procedures, and programs that shape the evolution and use of the Internet. No one person, company, organization or government runs the Internet. It is a globally distributed network comprising many voluntarily interconnected autonomous networks. It operates without a central governing body with each constituent network setting and enforcing its own policies. Its governance is conducted by a decentralized and international multistakeholder network of interconnected autonomous groups drawing from civil society, the private sector, governments, the academic and research communities and national and international organizations. They work cooperatively from their respective roles to create shared policies and standards that maintain the Internet's global interoperability for the public good.

Who-Runs-the-Internet-graphic

However, to help ensure interoperability, several key technical and policy aspects of the underlying core infrastructure and the principal namespaces are administered by the Internet Corporation for Assigned Names and Numbers (ICANN), which is headquartered in Los Angeles, California. ICANN oversees the assignment of globally unique identifiers on the Internet, including domain names, Internet protocol addresses, application port numbers in the transport protocols, and many other parameters. This seeks to create a globally unified namespace to ensure the global reach of the Internet. ICANN is governed by an international board of directors drawn from across the Internet's technical, business, academic, and other non-commercial communities. However, the National Telecommunications and Information Administration, an agency of the U.S. Department of Commerce, continues to have final approval over changes to the DNS root zone. This authority over the root zone file makes ICANN one of a few bodies with global, centralized influence over the otherwise distributed Internet. In the 30 September 2009 Affirmation of Commitments by the Department of Commerce and ICANN, the Department of Commerce finally affirmed that a "private coordinating process...is best able to flexibly meet the changing needs of the Internet and of Internet users". While ICANN itself interpreted this as a declaration of its independence, scholars still point out that this is not yet the case. Considering that the U.S. Department of Commerce can unilaterally terminate the Affirmation of Commitments with ICANN, the authority of DNS administration is likewise seen as revocable and derived from a single State, namely the United States.

The technical underpinning and standardization of the Internet's core protocols (IPv4 and IPv6) is an activity of the Internet Engineering Task Force (IETF), a non-profit organization of loosely affiliated international participants that anyone may associate with by contributing technical expertise.

On 16 November 2005, the United Nations-sponsored World Summit on the Information Society (WSIS), held in Tunis, established the Internet Governance Forum (IGF) to open an ongoing, non-binding conversation among multiple stakeholders about the future of Internet governance. Since WSIS, the term "Internet governance" has been broadened beyond narrow technical concerns to include a wider range of Internet-related policy issues. During 1979 the Internet Configuration Control Board was founded by DARPA to oversee the network's development. During 1984 it was renamed the Internet Advisory Board (IAB), and during 1986 it became the Internet Activities Board.

The Internet Engineering Task Force (IETF) was formed during 1986 by the U.S. government to develop and promote Internet standards. It consisted initially of researchers, but by the end of the year participation was available to anyone, and its business was performed largely by email.

From the early days of the network until his death during 1998, Jon Postel oversaw address allocation and other Internet protocol numbering and assignments in his capacity as Director of the Computer Networks

Division at the Information Sciences Institute of the University of Southern California, under a contract from the Department of Defense. This function eventually became known as the Internet Assigned Numbers Authority (IANA), and as it expanded to include management of the global Domain Name System (DNS) root servers, a small organization grew. Postel also served as RFC Editor.

Allocation of IP addresses was delegated to five Regional Internet Registries (RIRs):

1. American Registry for Internet Numbers (ARIN) for North America
2. Réseaux IP Européens - Network Coordination Centre (RIPE NCC) for Europe, the Middle East, and Central Asia
3. Asia-Pacific Network Information Centre (APNIC) for Asia and the Pacific region
4. Latin American and Caribbean Internet Addresses Registry (LACNIC) for Latin America and the Caribbean region
5. African Network Information Center (AfriNIC) was created in 2004 to manage allocations for Africa

After Jon Postel's death in 1998, IANA became part of ICANN, a California nonprofit established in September 1998 by the U.S. government and awarded a contract by the U.S. Department of Commerce. Initially two board members were elected by the Internet community at large, though this was changed by the rest of the board in 2002 in a poorly attended public meeting in Accra, Ghana.[24]

In 1992 the Internet Society (ISOC) was founded, with a mission to "assure the open development, evolution and use of the Internet for the benefit of all people throughout the world".[25] Its members include individuals (anyone may join) as well as corporations, organizations, governments, and universities. The IAB was renamed the Internet Architecture Board, and became part of ISOC. The Internet Engineering Task Force also became part of the ISOC. The IETF is overseen currently by the Internet Engineering Steering Group (IESG), and longer-term research is carried on by the Internet Research Task Force and overseen by the Internet Research Steering Group.

At the first World Summit on the Information Society in Geneva in 2003, the topic of Internet governance was discussed. ICANN's status as a private corporation under contract to the U.S. government created controversy among other governments, especially Brazil, China, South Africa, and some Arab states. Since no general agreement existed even on the definition of what comprised Internet governance, United Nations Secretary General Kofi Annan initiated a Working Group on Internet Governance (WGIG) to clarify the issues and report before the second part of the World Summit on the Information Society in Tunis 2005. After much controversial debate, during which the U.S. delegation refused to consider surrendering the U.S. control of the Root Zone file, participants agreed on a compromise to allow for wider international debate on the policy principles. They agreed to establish an Internet Governance Forum, to be convened by the United Nations Secretary General before the end of the second quarter of 2006. The Greek government volunteered to host the first such meeting.[26]

E-Government

E-government (short for electronic government, also known as e-gov, Internet government, digital government, online government, connected government) consists of the digital interactions between a citizen and their government (C2G), between governments and government agencies (G2G), between government and citizens (G2C), between government and employees (G2E), and between government and businesses/commerce (G2B).

Electronic government (or e-government) essentially refers to "The utilization of Information Technology (IT), Information and Communication Technologies (ICT s), and other web-based telecommunication technologies to improve and/or enhance on the efficiency and effectiveness of service delivery in the public sector." E-Government promotes and improves broad stakeholder's contribution to national and community development, as well as deepen the governance process.

E-government should enable anyone visiting a city website to communicate and interact with city employees via the Internet with graphical user interfaces (GUI), instant-messaging (IM), audio/video presentations, and in any way more sophisticated than a simple email letter to the address provided at the site" and "the use of technology to enhance the access to and delivery of government services to benefit citizens, business partners and employees". The focus should be on:

The use of information and communication technologies, and particularly the Internet, as a tool to achieve better government.

The use of information and communication technologies in all facets of the operations of a government organization.

The continuous optimization of service delivery, constituency participation and governance by transforming internal and external relationships through technology, the Internet and new media.

Whilst e-government has traditionally been understood as being centered on the operations of government, e-governance is understood to extend the scope by including citizen engagement and participation in governance. As such, following in line with the OECD definition of e-government, e-governance can be defined as the use of ICTs as a tool to achieve better governance.

Delivery models and activities of e-government

The primary delivery models of e-government can be divided into:

Government-to-citizen or government-to-consumer (G2C)

In this model, the G2C model apply the strategy of customer relationship management (CRM) with business concept.

By managing their customer (citizen) relationship, the business (government) can provide the needed products and services fulfill the needs from customer (citizen).

In United States, the NPR (National Partnership for Reinventing Government) has been implemented from 1993.

Government-to-business (G2B)

Government-to-government (G2G)

Government-to-employees (G2E)

Within each of these interaction domains, four kinds of activities take place:

1. Pushing information over the Internet, e.g.: regulatory services, general holidays, public hearing schedules, issue briefs, notifications, etc.
2. Two-way communications between the agency and the citizen, a business, or another government agency. In this model, users can engage in dialogue with agencies and post problems, comments, or requests to the agency.

3. Conducting transactions, e.g.: lodging tax returns, applying for services and grants.
4. governance, e.g.: To enable the citizen transition from passive information access to active citizen participation by:
 - a. Informing the citizen
 - b. Representing the citizen
 - c. Encouraging the citizen to vote
 - d. Consulting the citizen
 - e. Involving the citizen

Controversies of e-government

Disadvantages

The main disadvantages concerning e-government is the lack of equality in public access to the internet, reliability of information on the web, and hidden agendas of government groups that could influence and bias public opinions.

There are many considerations and potential implications of implementing and designing e-government, including disintermediation of the government and its citizens, impacts on economic, social, and political factors, vulnerability to cyber-attacks, and disturbances to the status quo in these areas

Hyper-surveillance

Increased contact between government and its citizens goes both ways. Once e-government begins to develop and become more sophisticated, citizens will be forced to interact electronically with the government on a larger scale. This could potentially lead to a lack of privacy for civilians as their government obtains more and more information on them. In a worst-case scenario, with so much information being passed electronically between government and civilians, a totalitarian-like system could develop. When the government has easy access to countless information on its citizens, personal privacy is lost.

Cost

Although "a prodigious amount of money has been spent" on the development and implementation of e-government, some say[who?] it has yielded only a mediocre product. The outcomes and effects of trial Internet-based governments are often difficult to gauge or unsatisfactory. According to Gartner, Worldwide IT spending is estimated to total \$3.6 trillion in 2011 which is 5.1% increase from the year 2010 (\$3.4 trillion).

Inaccessibility

An e-government site that provides web access and support often does not offer the "potential to reach many users including those who live in remote areas, are homebound, have low literacy levels, exist on poverty line incomes."

False sense of transparency and accountability

Opponents of e-government argue that online governmental transparency is dubious because it is maintained by the governments themselves. Information can be added or removed from the public eye. To this day, very few organizations monitor and provide accountability for these modifications. Those that do so, like the United States' OMBWatch[20] and Government Accountability Project, are often nonprofit

volunteers. Even the governments themselves do not always keep track of the information they insert and delete.

Advantages

The ultimate goal of the e-government is to be able to offer an increased portfolio of public services to citizens in an efficient and cost effective manner. E-government allows for government transparency. Government transparency is important because it allows the public to be informed about what the government is working on as well as the policies they are trying to implement. Simple tasks may be easier to perform through electronic government access. Many changes, such as marital status or address changes can be a long process and take a lot of paper work for citizens. E-government allows these tasks to be performed efficiently with more convenience to individuals. E-government is an easy way for the public to be more involved in political campaigns. It could increase voter awareness, which could lead to an increase in citizen participation in elections. It is convenient and cost-effective for businesses, and the public benefits by getting easy access to the most current information available without having to spend time, energy and money to get it.

E-government helps simplify processes and makes government information more easily accessible for public sector agencies and citizens. For example, the Indiana Bureau of Motor Vehicles simplified the process of certifying driver records to be admitted in county court proceedings. Indiana became the first state to allow government records to be digitally signed, legally certified and delivered electronically by using Electronic Postmark technology. In addition to its simplicity, e-democracy services can reduce costs. Alabama Department of Conservation & Natural Resources, Wal-Mart and NIC developed an online hunting and fishing license service utilizing an existing computer to automate the licensing process. More than 140,000 licenses were purchased at Wal-Mart stores during the first hunting season and the agency estimates it will save \$200,000 annually from service.

The anticipated benefits of e-government include efficiency, improved services, better accessibility of public services, sustainable community development and more transparency and accountability.

Democratization

One goal of e-government will be greater citizen participation. Through the internet, people from all over the country can interact with politicians or public servants and make their voices heard. Blogging and interactive surveys will allow politicians or public servants to see the views of the people they represent on any given issue. Chat rooms can place citizens in real-time contact with elected officials, their offices or provide them with the means to replace them by interacting directly with public servants, allowing voters to have a direct impact and influence in their government. These technologies can create a more transparent government, allowing voters to immediately see how and why their representation in the capital is voting the way they are. This helps voters better decide who to vote for in the future or how to help the public servants become more productive. A government could theoretically move more towards a true democracy with the proper application of e-government. Government transparency will give insight to the public on how decisions are made and hold elected officials or public servants accountable for their actions. The public could become a direct and prominent influence in government legislature to some degree.

Environmental bonuses

Proponents of e-government argue that online government services would lessen the need for hard copy forms. Due to recent pressures from environmentalist groups, the media, and the public, some governments and organizations have turned to the Internet to reduce this paper use. The United States government utilizes

the website <http://www.forms.gov> to provide “internal government forms for federal employees” and thus “produce significant savings in paper.

Speed, efficiency, and convenience

E-government allows citizens to interact with computers to achieve objectives at any time and any location, and eliminates the necessity for physical travel to government agents sitting behind desks and windows. Improved accounting and record keeping can be noted through computerization, and information and forms can be easily accessed, equaling quicker processing time. On the administrative side, access to help find or retrieve files and linked information can now be stored in databases versus hardcopies stored in various locations. Individuals with disabilities or conditions no longer have to be mobile to be active in government and can be in the comfort of their own homes.

Public approval

Recent trials of e-government have been met with acceptance and eagerness from the public. Citizens participate in online discussions of political issues with increasing frequency, and young people, who traditionally display minimal interest in government affairs, are drawn to electronic voting procedures.

Although internet-based governmental programs have been criticized for lack of reliable privacy policies, studies have shown that people value prosecution of offenders over personal confidentiality. Ninety percent of United States adults approve of Internet tracking systems of criminals, and 57% are willing to forgo some of their personal internet privacy if it leads to the prosecution of criminals or terrorists.

Technology-specific e-government

There are also some technology-specific sub-categories of e-government, such as m-government (mobile government), u-government (ubiquitous government), and g-government (GIS/GPS applications for e-government).

The primary delivery models of e-government are classified depending on who benefits. In the development of public sector or private sector portals and platforms, a system is created that benefits all constituents. Citizens needing to renew their vehicle registration have a convenient way to accomplish it while already engaged in meeting the regulatory inspection requirement. On behalf of a government partner, business provides what has traditionally, and solely, managed by government and can use this service to generate profit or attract new customers. Government agencies are relieved of the cost and complexity of having to process the transactions.

To develop these public sector portals or platforms, governments have the choice to internally develop and manage, outsource, or sign a self-funding contract. The self-funding model creates portals that pay for themselves through convenience fees for certain e-government transactions, known as self-funding portals.

Social networking services are an emerging area for e-democracy. The social networking entry point is within the citizens' environment and the engagement is on the citizens' terms. Proponents of e-government perceive government use of social networking as a medium to help government act more like the public it serves. Examples can be found at almost every state government portal through Facebook, Twitter, and YouTube widgets.